Exact recomputation of the layer margin across the mu-squared neighbourhood

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

ROBUSTNESS-MU2 remained OPEN because the layer margin m in the off-anchor STEP-5B re-margin (robustness-mu2-step5b-remargin v1.3) was held at its anchor value 0.00432 and only BOUNDED off-anchor, not recomputed. The closure bar (v1.3 §3, GATES.md) requires the exact $m(\mu^2)$ with $m(\mu^2) \geq 0.4\, m_{\text{anchor}}$ on $[\times 0.5, \times 2]$, the STEP-5B ratio recomputed with the exact margin, and a certified J_{eff} envelope. This note supplies all three. Scope: second-cumulant order, $\mu^2 \in [\times 0.5, \times 2] = [0.0025, 0.01]$ around the production anchor.

2. The layer margin is closed-form in mu-squared

The Prop-A layer margin (Math437 v1.2 / Math440) is the band penalty floor minus the band dip,

$$\text{MARGIN}(\mu^2) = P_B\big(M_+(\mu^2)\big) - \text{DIP}_{\text{BAND}}(\mu^2),$$

with (production convention, bare parameter $R = \mu^2$, gap point $r_R(\mu^2)$, $M_R(\mu^2) = M(r_R)$):

$$M_+(\mu^2) = \frac{-3u + \sqrt{9u^2 - 60v\,\mu^2}}{30v}, \qquad \hat{r}_0(M_c) = \mu^2 - \frac{3u^2}{20v}, \qquad \text{DIP}_{\text{BAND}} = \frac{|\hat{r}_0(M_c)|^{3/2}}{3\sqrt{v}},$$

$$P_B(M) = \tfrac{1}{2}(\mu^2 - r_R)(M - M_R) + \tfrac{3}{4}u\,(M^2 - M_R^2) + \tfrac{5}{2}v\,(M^3 - M_R^3).$$

At the anchor this reproduces the certified value exactly: $M_+ = 0.0510720$, $P_B(M_+) = 0.0052460$, $\text{DIP}_{\text{BAND}} = 0.0009260$, so $\text{MARGIN}(0.005) = 0.0043200$, validating the closed form against the frozen constant 0.00432.

3. The recomputation across the neighbourhood

Evaluating every μ^2-dependent input (r_R, M_R, M_+, $\hat{r}_0$) on a 61-point grid over $[0.0025, 0.01]$:

$$\boxed{\min_{[\times 0.5, \times 2]} m(\mu^2) = 0.004082 \text{ (at } \mu^2 = 0.0025\text{)}, \quad m(\mu^2) \geq 0.945\, m_{\text{anchor}}}$$

(max $= 0.004802$ at $\mu^2 = 0.01$; total drift 16.7% over the $\times 4$ band; $m(\mu^2)$ is smooth and monotone increasing in μ^2). The closure-bar condition $m(\mu^2) \geq 0.4\, m_{\text{anchor}}$ is met with large room (actual factor 0.945). The adjacent-grid Lipschitz remainder is 1.2×10^{-5}, so the grid minimum is a certified lower bound.

Recomputing the STEP-5B AddE ratio with the RECOMPUTED margin (not the frozen 0.00432):

μ^2	$m(\mu^2)$	ratio @ 4×10^{-4}	ratio @ 10^{-3}	ratio @ 2×10^{-3}
0.0025 ($\times$0.5)	0.004082	$\times$55.5	$\times$9.16	$\times$2.41
0.005 (anchor)	0.004320	$\times$59.4	$\times$9.80	$\times$2.58
0.01 ($\times$2)	0.004802	$\times$67.5	$\times$11.15	$\times$2.93

The worst ratio over the neighbourhood and all three intensities is $\times 2.41$ (the $\times 0.5$, $I = 2 \times 10^{-3}$ corner) – the STEP-5B closure holds with the EXACT margin, not the frozen anchor value. The earlier sampled-trend monotonicity is now a closed-form statement: $m(\mu^2)$ increases with μ^2, so the thinnest corner is the lower endpoint $\mu^2 = 0.0025$.

4. Certified J-eff envelope

At the thinnest corner the minimum-transfer envelope J_{eff} at quadrature resolutions $n_k = 500$ and $n_k = 1100$ agree to $< 0.1\%$ ($J_{\text{eff}} = 0.22354$ both); the endpoint ratio over this envelope stays > 1. The earlier $\times 9.8$ vs $\times 8.8$ discrepancy was a comparison across DIFFERENT margins, not a quadrature instability; at fixed inputs J_{eff} is converged.

5. Closure-bar status

All three closure-bar conditions are now met: (i) $m(\mu^2)$ recomputed exactly, $\geq 0.945\, m_{\text{anchor}}$; (ii) STEP-5B ratio > 1 ($\geq \times 2.41$) with the recomputed margin; (iii) certified J_{eff} envelope. The remaining honest qualifier is SCOPE: the whole analysis is second-cumulant order (SC-SCOPE). Subject to operator authorization, ROBUSTNESS-MU2 is ready to be marked CLOSED@[$\times$0.5, $\times$2]-2ND-CUMULANT. This note does NOT itself flip the gate (operator authorizes gate transitions).

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"The closed form is just the frozen constant in disguise."* DISMISSED: every μ^2-dependent input is recomputed; the margin genuinely VARIES (0.004082–0.004802, a 16.7% swing), and the thinnest corner shifts to $\mu^2 = 0.0025$, lowering the endpoint ratio from the frozen $\times 2.55$ to the exact $\times 2.41$. The anchor value 0.00432 is reproduced as the $\mu^2 = 0.005$ special case.

(β) *"A 61-point grid is not an interval proof."* VALID-with-mitigation: $m(\mu^2)$ is C^1 and monotone; the adjacent-grid variation is 1.2×10^{-5}, three orders below the margin, so the grid minimum is a certified lower bound to that remainder. The closure verdict has a $> \times 2.4$ safety factor, far exceeding any grid remainder.

(γ) *"This closes ROBUSTNESS-MU2 unconditionally."* DISMISSED: the gate is CLOSED only at second-cumulant scope and only on [$\times 0.5, \times 2$]; both qualifiers are stated. The flip itself awaits operator authorization.

Quantitative sanity check (mandatory): magnitude – MARGIN(0.005) = 0.0043200 reproduces the certified 0.00432 to 5 figures; limit-case – the two-branch regime requires $9u^2 - 60v\mu^2 > 0$, i.e. $\mu^2 < 0.0342$, so $[0.0025, 0.01]$ is safely inside; sign-direction – $m(\mu^2)$ increasing in μ^2 makes the lower endpoint the worst corner, consistent with the recomputed table.

7. Result footer

Result ID:	B1-RH-ENUM / ROBUSTNESS-MU2 exact margin recomputation
Precise statement:	MARGIN(mu^2) = PB(M_+(mu^2)) - DIP_BAND(mu^2), closed-form; reproduces 0.00432 at the anchor; over [x0.5,x2] min m(mu^2) = 0.004082 = 0.945 m_anchor (>= 0.4 m_anchor). STEP-5B ratio with the recomputed margin: worst x2.41 > 1. J_eff envelope converged.
Scope:	Second-cumulant order; mu^2 in [0.0025, 0.01].
Dependencies:	Prop-A band floor (Math437 v1.2/Math440); production gap_solve/M_fast; AddE de-thinned ratios.
Evidence grade:	EXECUTED (robustness_mu2_margin_recompute.py, 11/11 asserts) + closed-form anchor validation.
Reproduction command:	python codes/vacuum/robustness_mu2_margin_recompute.py
Expected output:	11/11 PASS; MARGIN(0.005)=0.00432; min m(mu^2)=0.004082 (0.945 anchor); worst STEP-5B ratio x2.41.
Falsification gate:	m(mu^2) < 0.4 m_anchor anywhere on [x0.5,x2]; the STEP-5B ratio with the recomputed margin <= 1.
Tier before / after:	No tier change. ROBUSTNESS-MU2 closure bar MET; CLOSE@[x0.5,x2]-2ND-CUMULANT RECOMMENDED pending operator sign-off. Gate stays OPEN until then.
No-overclaim statement:	Closure is second-cumulant scope and [x0.5,x2] only. The note supplies the missing exact m(mu^2); it does not itself flip the gate. All-orders robustness is SC-SCOPE (separate).
Next required action:	operator authorization to flip ROBUSTNESS-MU2 -> CLOSED@[x0.5,x2]-2ND-CUMULANT (atomic GATES.md + card + CHANGELOG).